Type MS · Exponential equations, including hidden quadratics

Answer 1 of 7 · Question 11 part a.ii · 3 marks

2023 JAN · Paper 02 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Exponential equations, including hidden quadratics

Answer 2 of 7 · Question 11 part a.ii · 3 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a.ii in this mark scheme extract

Number		
11 (a)(i)	$(a=)-8$	B1 [1]
(a)(ii)	$e^{2x}=9\Rightarrow 2x=\ln 9$	M1
	$x=\frac{1}{2}\ln 9=\ln 3$ *	A1cso [2]
ALT	$e^{2x}=9\Rightarrow\left[\left(e^{x}\right)^{2}=3^{2}\right]\Rightarrow e^{x}=3$	M1
	$x=\ln 3$ *	A1cso
(b)	$\pi\int_{0}^{\ln 3}\left(e^{2x}-9\right)^{2}dx$	M1
	$\pi\int_{0}^{\ln 3}\left(e^{4x}-18e^{2x}+81\right)dx$	A1
	$(\pi)\left[\frac{1}{4}e^{4x}-9e^{2x}+81x\right]_{(0)}^{("\ln 3")}$ oe	M1 A1ft
	$(\pi)\left[\left(\frac{81}{4}-81+81\ln 3\right)-\left(\frac{1}{4}-9\right)\right]$	M1
	$\pi(81\ln 3-52)$	A1 [6]
		Total 9 marks

Type MS · Exponential equations, including hidden quadratics

Answer 2 of 7 · Question 11 part a.ii · 3 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a.ii in this mark scheme extract

π must only be present for the final A1 mark and the first M1 mark (can be implied by first or final A1)

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Type MS · Exponential equations, including hidden quadratics

Answer 2 of 7 · Question 11 part a.ii · 3 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a.ii in this mark scheme extract

Type MS · Exponential equations, including hidden quadratics

Answer 3 of 7 · Question 10 part a · 4 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

10(a)	$\frac{1}{243}=3^{-5}$	B1
	$9^{3y}=3^{6y}$	B1
	$\frac{9^{3y}}{243}=3^{-5}\times 3^{6y} \Rightarrow \frac{9^{3y}}{243}=3^{(6y-5)}$ *	M1A1 cso [4]
(b)	$27^{(x-2)}=3^{3(x-2)}=3^{(3x-6)}$	M1
	$6y-5=3x-6 \Rightarrow (6y-3x-1=0)$	M1
	$\log_4 2=\frac{1}{2}$	B1
	$\log_{10}\sqrt{6xy}=\frac{1}{2}\log_{10}(6xy) \Rightarrow \log_{10}(6xy)=1$	M1
	$1=\log_{10}10 \Rightarrow \log_{10}(6xy)=\log_{10}10 \Rightarrow 6xy=10$	M1
	$3x-6y-1=0$ $6xy=10$	
	Method A	
	$6y=\frac{10}{x} \Rightarrow 3x-\frac{10}{x}-1=0 \Rightarrow 3x^2-x-10=0$	M1
	$3x^2-x-10=(3x+5)(x-2)=0 \Rightarrow x=2,\ -\frac{5}{3}$	M1
	$3\times 2-6y-1=0 \Rightarrow 6y=5 \Rightarrow y=\frac{5}{6}$	A1
	$3\times\left(-\frac{5}{3}\right)-6y-1=0 \Rightarrow -6y=6 \Rightarrow y=-1$	
	$x=2 \quad y=\frac{5}{6}$ or $x=-\frac{5}{3} \quad y=-1$	A1 [9]
	Method B	
	$3x=\frac{5}{y} \Rightarrow \frac{5}{y}-6y-1=0 \Rightarrow 6y^2+y-5=0$	M1
	$6y^2+y-5=(6y-5)(y+1)=0 \Rightarrow y=\frac{5}{6},\ -1$	M1
	$3x-6\times\frac{5}{6}-1=0 \Rightarrow 3x=6 \Rightarrow x=2$	A1
	$3x-6\times(-1)-1=0 \Rightarrow 3x=-5 \Rightarrow x=-\frac{5}{3}$	
	$x=2 \quad y=\frac{5}{6}$ or $x=-\frac{5}{3} \quad y=-1$	A1 [9]
		Total 13 marks

Type MS · Exponential equations, including hidden quadratics

Answer 3 of 7 · Question 10 part a · 4 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
10	$\dfrac{9^{3y}}{243}=27^{(x-2)}$ $\log_{10}\sqrt{6xy}=\log_4 2$	
(a)	Writes down $\dfrac{1}{243}=3^{-5}$ May be seen as $\frac{1}{243}=\frac{1}{3^5}$ later correctly used as 3^{-5} when combining terms.	B1
	Writes down $9^{3y}=3^{6y}$	B1
	Combines the terms $\dfrac{9^{3y}}{243}=3^{-5}\times 3^{6y}=3^{(6y-5)}$	M1
	For the correct expression with no errors. $\dfrac{9^{3y}}{243}=3^{(6y-5)}*$	A1 cso [4]
(b)	For dealing with the power of 3 to give $27^{(x-2)}=3^{3(x-2)}=3^{(3x-6)}$	M1
	For equating the powers of 3 to give the equation $6y-5=3x-6\Rightarrow(6y-3x-1=0)$	M1
	For stating $\log_4 2=\dfrac{1}{2}$	B1
	For dealing with the square root $\log_{10}\sqrt{6xy}=\dfrac{1}{2}\log_{10}(6xy)=\left(\dfrac{1}{2}\right)\Rightarrow\log_{10}(6xy)=1$	M1
	For correctly removing all logarithms from the second equation $1=\log_{10}10\Rightarrow\log_{10}(6xy)=\log_{10}10\Rightarrow 6xy=10$	M1
$3x-6y-1=0$ $6xy=10$		
	Method A	
	For substituting $6y=\dfrac{10}{x}$ into the linear equation to give $3x-\dfrac{10}{x}-1=0$ and attempting to form a 3TQ $3x^2-x-10=0$ OR for substituting $y=\frac{1}{6}(3x-1)$ into $6xy=10$ to give $x(3x-1)=10$ and attempting to form a 3TQ $3x^2-x-10=0$	M1
	For attempting to solve their 3TQ $3x^2-x-10=(3x+5)(x-2)=0\Rightarrow x=2,\ -\dfrac{5}{3}$	M1

Type MS · Exponential equations, including hidden quadratics

Answer 3 of 7 · Question 10 part a · 4 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

	For finding the values of y $3\times 2-6y-1=0\Rightarrow 6y=5\Rightarrow y=\frac{5}{6}$ $3\times\left(-\frac{5}{3}\right)-6y-1=0\Rightarrow -6y=6\Rightarrow y=-1$ $x=2\quad y=\frac{5}{6}$ or $x=-\frac{5}{3}\quad y=-1$ Answers must be given in pairs, pairing may be implied from working.	A1 A1 [9]
	Method B	
	For substituting $3x=\frac{10}{2y}\Rightarrow 3x=\frac{5}{y}$ into the linear equation to give $\frac{5}{y}-6y-1=0$ and attempting to form a 3TQ $6y^2+y-5=0$ OR for substituting $x=\frac{1}{3}(6y+1)$ into $6xy=10$ to give $2(6y+1)y=10$ and attempting to form a 3TQ $6y^2+y-5=0$	M1
	For attempting to solve their 3TQ $6y^2+y-5=(6y-5)(y+1)=0\Rightarrow y=\frac{5}{6},\ -1$	M1
	For finding the values of x $3x-6\times\frac{5}{6}-1=0\Rightarrow 3x=6\Rightarrow x=2$ $3x-6\times(-1)-1=0\Rightarrow 3x=-5\Rightarrow x=-\frac{5}{3}$ $x=2\quad y=\frac{5}{6}$ or $x=-\frac{5}{3}\quad y=-1$	A1 A1 [9]
		Total 13 marks

Type MS · Exponential equations, including hidden quadratics

Answer 3 of 7 · Question 10 part a · 4 marks

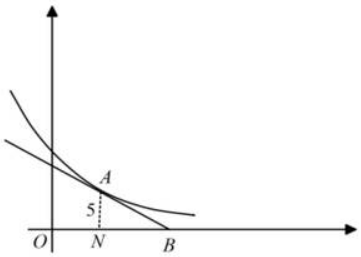
2023 JAN · Paper 02 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Exponential equations, including hidden quadratics

Answer 4 of 7 · Question 8 part a · 4 marks

2020 NOV · Paper 02 · extract 1 of 4 · find part a in this mark scheme extract

8(a)	$5e^{-2x}+4=e^{2x}$ $\quad 5e^{-2x}+4-e^{2x}=0$	OR $\quad y=\frac{5}{y}+4\Rightarrow y^2-4y-5=0$	M1
	$\left(5e^{-x}-e^{x}\right)\left(e^{-x}+e^{x}\right)=0$	$(y-5)(y+1)=0$	M1
	$5e^{-x}=e^{x}$ $\quad e^{2x}=5$ $\quad x=\frac{1}{2}\ln 5$ $\left(\text{oe eg } \ln\sqrt{5}\right)$	$y=5$	A1
	$\left(e^{-x}=-e^{x}\right.$ not possible$)$	$e^{2x}=5$ $\quad x=\frac{1}{2}\ln 5$	
	A is $\left(\frac{1}{2}\ln 5, 5\right)$		A1 (4)
(b)	$y=5e^{-2x}+4\Rightarrow \frac{dy}{dx}=-10e^{-2x}$		M1
	At A $\quad \frac{dy}{dx}=-10e^{-2x}=-10\times\frac{1}{5}=-2$		A1ft
	Eqn tgt: $\quad y-5=-2\left(x-\frac{1}{2}\ln 5\right)$		dM1A1
	$y=0\Rightarrow x=\frac{1}{2}(5+\ln 5)$ $\quad (=x$ coordinate of $B)$*		A1cso (5)
ALT	For last 3 marks:		
	Hence $\frac{5}{NB}=2\Rightarrow NB=\frac{5}{2}$		dM1A1
	$ON=\frac{1}{2}\ln 5$		
	$OB=\frac{1}{2}\ln 5+\frac{5}{2}=\frac{1}{2}\ 5+\ln 5$ *		A1cso
(c)	$C_2: \frac{dy}{dx}=2e^{2x}\Rightarrow$ grad tgt at A is $2\times 5=10$		B1ft
	Eqn tgt: $\quad y-5=10\left(x-\frac{1}{2}\ln 5\right)$		M1
	At D: $\quad x=\frac{1}{2}(-1+\ln 5)$		A1
	Area $\triangle ABD=\frac{1}{2}\left(\frac{1}{2}(5+\ln 5)-\frac{1}{2}(-1+\ln 5)\right)\times 5$		M1A1
	$=\frac{15}{2}$ or $7\frac{1}{2}$ $\left(\text{units}^2\right)$		A1 (6)
	See notes for area by "determinant" method		

Type MS · Exponential equations, including hidden quadratics

Answer 4 of 7 · Question 8 part a · 4 marks

2020 NOV · Paper 02 · extract 2 of 4 · find part a in this mark scheme extract

ALT	For second and third marks: $\frac{5}{ND} = 10 \Rightarrow ND = \frac{1}{2}$ $OD = \frac{1}{2}\ln 5 - \frac{1}{2}$	 M1 A1 [15]
(a)		
M1	Equate the 2 curve equations. No need to simplify	
M1	Factorise their equation	
A1	Obtain the one possible value for x (other root need not be seen; if seen it must be rejected) **Must be exact**	
A1	Obtain the corresponding value for y. **Must be exact.** Need not be shown in coordinate brackets. Use of $e^{2x} = 5$ leads to $y = 5$ without use of a value of x, so M1M1A0A1 can be scored. There must only be one correct y shown. Accept $y = e^{\ln 5}$	
(b)		
M1	Differentiate the equation of C_1 $5e^{-2x} \rightarrow ke^{-2x}$ where $k = \pm 5$ or ± 10 and no integration seen	
A1ft	Grad at A = -2 follow through their x coordinate	
dM1	Obtain the equation of the tangent at A using their gradient and their coordinates of A. Can be in any form but if $y = mx + c$ is used a value for c must be found. Gradient of the tangent must be numerical.	
A1	Correct equation in any form	
A1cso	Correct x coordinate of B obtained from correct working.	
ALT	For last 3 marks	
dM1	Use their y coordinate of A and their (numerical) gradient of the tangent to find the length NB (where N is the foot of the perpendicular from A to the x-axis)	
A1	Correct length of NB	
A1cso	Add the x coordinate of A to obtain the x coordinate of B	
(c)		
B1ft	Correct gradient of tangent to C_2 at A follow through their x coordinate	
M1	Obtain an equation for the tangent using their gradient and their coordinates of A Gradient of the tangent must be numerical.	
A1	Correct x coordinate of D (exact or minimum 3 sf)	
M1	Use a correct formula for the area of a triangle with their y coordinate of A, their x coordinate of D and the **given** x coordinate of B	
A1	Correct, unsimplified area Allow use of correct but non-exact coordinates	
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5	
	Heron's formula: Nos which may be seen: $AB = \frac{5\sqrt{5}}{2}, AD = \frac{\sqrt{101}}{2}, BD = 3, s = \frac{1}{2}(a+b+c) = 6.8$	
ALT	For second and third marks:	
M1	Use their y coordinate of A and their gradient of the tangent to find the length ND	
A1	Use the x coordinate of A to obtain the x coordinate of D	

Type MS · Exponential equations, including hidden quadratics

Answer 4 of 7 · Question 8 part a · 4 marks

2020 NOV · Paper 02 · extract 3 of 4 · find part a in this mark scheme extract

ALT	Area by "determinant" method:
M1	Eg Area $= \frac{1}{2}\begin{vmatrix} \frac{1}{2}\ln 5 & \frac{1}{2}(5+\ln 5) & \frac{1}{2}(\ln 5-1) & \frac{1}{2}\ln 5 \\ 5 & 0 & 0 & 5 \end{vmatrix}$ y coordinates of B and D must be 0 Must include the ½ and have 4 sets of coordinates with first and last the same.
A1	$= \frac{1}{2}\left(\frac{1}{2}(\ln 5-1)\times 5 - \frac{1}{2}(\ln 5+1)\right)$ Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5 **Must be positive**

Type MS · Exponential equations, including hidden quadratics

Answer 4 of 7 · Question 8 part a · 4 marks

2020 NOV · Paper 02 · extract 4 of 4 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$2+4x-x^2$	

Type MS · Exponential equations, including hidden quadratics

Answer 5 of 7 · Question 8 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a.i in this mark scheme extract

8 (a)(i)	$e^{3x} - 1 = 9 - 9e^{-3x}$	M1
	$(e^{3x})^2 - e^{3x} = 9e^{3x} - 9$	M1
	$\left(e^{3x}\right)^2 - 10e^{3x} + 9 = 0$ *	A1cso*
(ii)	$(e^{3x} - 1)(e^{3x} - 9)$ [= 0] leading to $e^{3x} =$	M1
	$e^{3x} = 9 \quad \rightarrow \quad x = \frac{1}{3}\ln 9$ *	A1cso* [5]
Note: subscripts on marks in (b) indicate which mark is being awarded on epen (1=1st etc)		
(b)	$\left(\int_0^{\frac{1}{3}\ln 9}(9 - 9e^{-3x})\,dx =\right)$ $9x + 3e^{-3x}$	$M1_1$ $A1_2A1_3$
	$\left(\int_0^{\frac{1}{3}\ln 9}(e^{3x} - 1)\,dx =\right)$ $\frac{e^{3x}}{3} - x$	
	"$9x + 3e^{-3x}$" − "$\frac{e^{3x}}{3} - x$" $M1_4$ $\left(10 \times \frac{1}{3}\ln 9 - \frac{9}{-3}e^{-3\times\frac{1}{3}\ln 9} - \frac{e^{3\times\frac{1}{3}\ln 9}}{3}\right) - \left(10 \times 0 - \frac{9}{-3}e^{-3\times 0} - \frac{e^{3\times 0}}{3}\right)$ $dM1_5$	$\left(9 \times \frac{1}{3}\ln 9 + 3e^{-3\times\frac{1}{3}\ln 9}\right) - (9 \times 0 + 3e^0)$ $dM1_5$ **and** $\left(\frac{e^{3\times\frac{1}{3}\ln 9}}{3} - \frac{1}{3}\ln 9\right) - \left(\frac{e^0}{3} - 0\right)$ "$3\ln 9 - \frac{8}{3}$"−$\left("\frac{8}{3} - \frac{1}{3}\ln 9"\right)$ $M1_4$
	$\frac{10}{3}\ln 9 - \frac{16}{3}$	$A1_6$
	ALT $\int(9 - 9e^{-3x}) - (e^{3x} - 1)dx$	$M1_4$
	$(\int(10 - 9e^{-3x} - e^{3x})\,dx =)\ 10x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3}$	$M1_1$ $A1_2\ A1_3$
	$\left(10\times\frac{1}{3}\ln 9 - \frac{9}{-3}e^{-3\times\frac{1}{3}\ln 9} - \frac{e^{3\times\frac{1}{3}\ln 9}}{3}\right) - \left(10\times 0 - \frac{9}{-3}e^{-3\times 0} - \frac{e^{3\times 0}}{3}\right)$	$dM1_5$
	$\frac{10}{3}\ln 9 - \frac{16}{3}$ oe	$A1_6$ [6]
		Total 11 marks

Type MS · Exponential equations, including hidden quadratics

Answer 5 of 7 · Question 8 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a.i in this mark scheme extract

Part	Mark	Additional Guidance
Mark parts (i) and (ii) together.		
(a)(i)	M1	For equating the two equations.
	M1	For multiplying through by e^{3x}, minimum of 2 out of 4 correct terms. (presence of $\pm 10e^{3x}$ indicates 2 correct terms).
	A1*cso	Correct solution only, no errors or omissions.
(ii)	M1	Minimally acceptable attempt at solving the equation leading to $e^{3x} =$ See general guidance, if the formula is quoted allow up to two slips in substitution, otherwise the substitution must be correct.
	A1*cso	Correct solution only, no errors or omissions. If 0 also included then this should be rejected.
(b)	$M1_1$	For attempt to integrate one of: $9 - 9e^{-3x}$ or $e^{3x} - 1$ or $\pm[(9 - 9e^{-3x}) - (e^{3x} - 1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	$A1_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	$A1_3$	For correct integration of both curves $9x + 3e^{-3x}$ **and** $\frac{e^{3x}}{3} - x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3}\right)$ or $\pm\left(9x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3} + x\right)$ **Note: this is an M mark in epen**
	$M1_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	$dM1_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	$A1_6$	For the correct answer oe. Must be exact value.

Type MS · Exponential equations, including hidden quadratics

Answer 5 of 7 · Question 8 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Exponential equations, including hidden quadratics

Answer 6 of 7 · Question 8 part a.ii · 5 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a.ii in this mark scheme extract

8 (a)(i)	$e^{3x} - 1 = 9 - 9e^{-3x}$	M1
	$(e^{3x})^2 - e^{3x} = 9e^{3x} - 9$	M1
	$\left(e^{3x}\right)^2 - 10e^{3x} + 9 = 0$ *	A1cso*
(ii)	$(e^{3x} - 1)(e^{3x} - 9)$ [= 0] leading to $e^{3x} =$	M1
	$e^{3x} = 9 \quad \rightarrow \quad x = \frac{1}{3}\ln 9$ *	A1cso* [5]
Note: subscripts on marks in (b) indicate which mark is being awarded on epen (1=1st etc)		
(b)	$\left(\int_0^{\frac{1}{3}\ln 9}(9 - 9e^{-3x})\,dx =\right)$ $9x + 3e^{-3x}$	$M1_1$ $A1_2A1_3$
	$\left(\int_0^{\frac{1}{3}\ln 9}(e^{3x} - 1)\,dx =\right)$ $\frac{e^{3x}}{3} - x$	
	"$9x + 3e^{-3x}$" − "$\frac{e^{3x}}{3} - x$" $M1_4$ $\left(10 \times \frac{1}{3}\ln 9 - \frac{9}{-3}e^{-3\times\frac{1}{3}\ln 9} - \frac{e^{3\times\frac{1}{3}\ln 9}}{3}\right) - \left(10 \times 0 - \frac{9}{-3}e^{-3\times 0} - \frac{e^{3\times 0}}{3}\right)$ $dM1_5$	$\left(9 \times \frac{1}{3}\ln 9 + 3e^{-3\times\frac{1}{3}\ln 9}\right) - (9 \times 0 + 3e^0)$ $dM1_5$ **and** $\left(\frac{e^{3\times\frac{1}{3}\ln 9}}{3} - \frac{1}{3}\ln 9\right) - \left(\frac{e^0}{3} - 0\right)$ "$3\ln 9 - \frac{8}{3}$"−$\left("\frac{8}{3} - \frac{1}{3}\ln 9"\right)$ $M1_4$
	$\frac{10}{3}\ln 9 - \frac{16}{3}$	$A1_6$
	ALT $\int(9 - 9e^{-3x}) - (e^{3x} - 1)dx$	$M1_4$
	$(\int(10 - 9e^{-3x} - e^{3x})\,dx =)\ 10x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3}$	$M1_1$ $A1_2\ A1_3$
	$\left(10\times\frac{1}{3}\ln 9 - \frac{9}{-3}e^{-3\times\frac{1}{3}\ln 9} - \frac{e^{3\times\frac{1}{3}\ln 9}}{3}\right) - \left(10\times 0 - \frac{9}{-3}e^{-3\times 0} - \frac{e^{3\times 0}}{3}\right)$	$dM1_5$
	$\frac{10}{3}\ln 9 - \frac{16}{3}$ oe	$A1_6$ [6]
		Total 11 marks

Type MS · Exponential equations, including hidden quadratics

Answer 6 of 7 · Question 8 part a.ii · 5 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a.ii in this mark scheme extract

Part	Mark	Additional Guidance
Mark parts (i) and (ii) together.		
(a)(i)	M1	For equating the two equations.
	M1	For multiplying through by e^{3x}, minimum of 2 out of 4 correct terms. (presence of $\pm 10e^{3x}$ indicates 2 correct terms).
	A1*cso	Correct solution only, no errors or omissions.
(ii)	M1	Minimally acceptable attempt at solving the equation leading to $e^{3x} =$ See general guidance, if the formula is quoted allow up to two slips in substitution, otherwise the substitution must be correct.
	A1*cso	Correct solution only, no errors or omissions. If 0 also included then this should be rejected.
(b)	$M1_1$	For attempt to integrate one of: $9 - 9e^{-3x}$ or $e^{3x} - 1$ or $\pm[(9 - 9e^{-3x}) - (e^{3x} - 1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	$A1_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	$A1_3$	For correct integration of both curves $9x + 3e^{-3x}$ **and** $\frac{e^{3x}}{3} - x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3}\right)$ or $\pm\left(9x - \frac{9}{-3}e^{-3x} - \frac{e^{3x}}{3} + x\right)$ **Note: this is an M mark in epen**
	$M1_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	$dM1_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	$A1_6$	For the correct answer oe. Must be exact value.

Type MS · Exponential equations, including hidden quadratics

Answer 6 of 7 · Question 8 part a.ii · 5 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Exponential equations, including hidden quadratics

Answer 7 of 7 · Question 5 part c · 5 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks
5(a)	$f(-2)=-16a+4+2b+3a=0$ and $f(1)=2a+1-b+3a=0$	M1M1
	$-13a+4+2b=0$ and $5a+1-b=0 \Rightarrow a=2^*$ $b=11$	M1A1cso A1 [5]
(b)	$f(x)=4x^3+x^2-11x+6=(4x-3)(x+2)(x-1)$	M1A1 [2]
(c)	$h(y)=2^{(3y+2)}+2^{2y}-11(2^y)+6 \Rightarrow h(y)=4(2^y)^3+(2^y)^2-11(2^y)+6$	M1
	$x=2^y \Rightarrow 2^y-1=0,\ 2^y+2=0,\ 4(2^y)-3=0$	M1
	$2^y=1 \Rightarrow y=0$	B1
	$2^y=\frac{3}{4} \Rightarrow y=\log_2\frac{3}{4}=-0.4150... \Rightarrow y=-0.415$ $[2^y=-2$ no solution$]$	M1A1 [5]
		Total 12 marks

Question	Notes	Marks
(a)	For substituting ± 2 into $f(x)=0$ Allow sign errors	M1
	For substituting ± 1 into $f(x)=0$ Allow sign errors	M1
	For solving the resulting pair of simultaneous equations. $-13a+4+2b=0$ $5a+1-b=0$	M1
	For $a=2$ *	A1 cso
	For $b=11$	A1 [5]
	ALT uses polynomial division.	
	Multiplies out the product of the two factors to give: $(x+2)(x-1)=x^2+x-2$ This must be correct.	M1

Type MS · Exponential equations, including hidden quadratics

Answer 7 of 7 · Question 5 part c · 5 marks

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	Uses polynomial division to give as a minimum: $2ax+(1-2M)$ where M is a constant. $x^2+x-2\overline{)2ax^3+x^2-bx+3a}$ $\quad 2ax+(1-2a)$	M1
	Equates coefficients: $(x^2+x-2)(2ax+(1-2a))=2ax^3+x^2-bx+3a$ $\Rightarrow 2ax^3+x^2+x(1-6a)+(-2+4a)=2ax^3+x^2-bx+3a$ $3a=4a-2$ $1-6a=-b$	M1
	For $a=2$ *	A1 cso
	For $b=11$	A1
(b)	Divides $f(x)=4x^3+x^2-11x+6$ by either $[x^2+x-2]$ or $x+2$ or $x-1$ Either, $\left[x^2+x-2\overline{)4x^3+x^2-11x+6}\right.$ $\quad 4x+k$ $x+2\overline{)4x^3+x^2-11x+6}$ $\quad 4x^2-7x\pm k$ $x-1\overline{)4x^3+x^2-11x+6}$ $\quad 4x^2+5x\pm k$	M1
	For the correct factorisation of $f(x)$ $4x^2-7x+3=(4x-3)(x-1)\Rightarrow f(x)=(4x-3)(x-1)(x+2)$	A1 [2]
(c)	For manipulating the indices to achieve $\Rightarrow h(y)=4(2^y)^3+(2^y)^2-11(2^y)+6$ OR Uses the factorised expression to obtain: $h(y)=(2^y+2)(2^y-1)(4(2^y)-3)$	M1
	Substitutes $x=2^y$ into the factorised f (x) to find either $2^y-1=0$ or $2^y+2=0$ or $4(2^y)-3=0$	M1
	For $2^y=1\Rightarrow y=0$	B1
	For $4(2^y)=3\Rightarrow 2^y=\frac{3}{4}\Rightarrow y=\log_2\frac{3}{4}$	M1
	For awrt $y=-0.4150...\Rightarrow y\approx -0.415$ [No solution for $2^y=-2$ - must reject]	A1 [5]
		Total 12 marks

Type MS · Exponential equations, including hidden quadratics

Answer 7 of 7 · Question 5 part c · 5 marks

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Question	Scheme	Marks